Documentation — Phase 1: WHAT

Purpose. Define the solution architecture for **Aviation — U.S. Flight Delay Network**. No code yet. The mentor reviews and signs off this document before implementation begins.

Team

Role	Name
Team lead	Akhmedova Muslima
Member 2	O'ralov Ilyos
Member 3	Aliqulov Davron
Member 4	Gulamov Abdulaziz
Member 5	Boymuhamedov Komronbek

Date: Chosen scenario: Aviation — U.S. Flight Delay Network

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1. Problem Restatement

In **your own words**, restate the business problem in 3–5 sentences.

We need to analyze data from more than 5.8 million flights in the United States and create a network of connections between airports. Our main goal is to identify the 5 most "infectious" airports that start a chain of delays. Through this system, we will show how a malfunction at one airport affects air traffic throughout the country (spillover effect).

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2. Proposed Solution Stack

List the **specific tools, libraries, and methods** you plan to use at each stage.

2.1 Data download & ingestion

- *Kaggle API: To pull the US Flights 2015 dataset directly into the environment.*
- *Pandas: For high-performance CSV reading, utilizing dtype optimization to handle 5.8M rows.*
- *Storage: Data will be stored locally*

2.2 Data cleaning / hygiene

- *Remove cancelled/diverted flights — to avoid biased delay analysis → pandas filtering*
- *Handle missing values — to ensure clean calculations → dropna() / fillna()*
- *Convert datetime columns — to enable time-based analysis → pandas.to_datetime*

- *Remove duplicates — to avoid double counting → drop_duplicates()*
- *Select relevant features — to simplify analysis → column filtering*

2.3 Analysis / modelling

• Method: Directed Weighted Graph Analysis using NetworkX.

• Graph Construction:

- Nodes: Unique Airports (Origin/Destination).

- Edges: Directed flight routes (Origin → Destination).

- Edge Weight: The sum of Arrival_Delay minutes for all flights on that route where Departure_Delay > 15 minutes. This explicitly quantifies "Delay Spillover Volume."

• Centrality Metric: Weighted Out-Strength Centrality.

- Calculation: For each airport, sum the weights of all outgoing edges.

- Interpretation: This score represents the total minutes of delay an airport propagates to the rest of the network. High scores indicate "Super-Spreader" airports.

• Ranking: Sort all nodes by Weighted Out-Strength in descending order to identify the Top 5 most infectious airports.

2.4 Visualization

- *Bar chart — to show the Top 5 "infectious" airports and compare their delay impact*
- *Network graph — to visualize how airports are connected and how delays spread*
- *Heatmap — to show delay spillover between origin and destination airports*
- *Time series plot — to analyze how delays change over time*

2.5 Reporting

- *Google colab link*

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3. Logic Map

*Describe the data flow in stages. Each stage = **input → transformation → output**.*

Stage 1 — Ingestion

Input: Raw flights.csv from Kaggle (5.8M+ rows).
Action: Load into Pandas DataFrame; verify schema.
Output: Raw DataFrame in memory.

Stage 2 — Cleaning
Input: Raw DataFrame.
Action: Select columns → Drop high-null cols → Filter negative delays → Convert time formats → Lowercase headers.

Output: cleaned_flights_2015.csv (Hygienic, standardized dataset).

Stage 3 — Analysis
Input: Cleaned CSV
Action: Filter for significant delays (>15m) → Group by Route (Origin/Dest) → Sum Delay Minutes → Build DiGraph → Calculate Weighted Out-Degree.
Output: Network Graph Object (G) and a sorted list of Airport Scores.

Stage 4 — Visualization
Input: Network Graph G and Top 5 List
Action: Compute layout positions → Map scores to node sizes/colors → Render edges → Annotate Top 5 labels.
Output: network_chart.png (Visual representation of delay propagation)

Stage 5 — Insight & Recommendation
Input: Top 5 List + Network Chart
Action: Synthesize findings into a narrative explaining why these airports are critical and propose buffer-time strategies.
Output: written deliverable

Replace each `...` with concrete details for **Aviation — U.S. Flight Delay Network**.

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4. Workload Split (5 Developers)

Developer	Responsibility	Stages owned
1 Gulamov Abdulaziz	Dataset setup	Stage 1
2 Aliqulov Davron	Data cleaning	Stage 2
3 Boymuhamedov Komronbek	Graph analysis	Stage 3
4 O'ralov Ilyos	Visualization	Stage 4
5 Akhmedova Muslima	Reporting & documentation	Stage 5

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5. Risks & Mitigation

List 3 things that could go wrong and your mitigation plan.

1. *Large dataset may cause memory/performance issues — Use Pandas dtype optimization and process only relevant columns.*
2. *Missing or inconsistent delay values may affect analysis accuracy — Apply data cleaning, filtering, and missing value handling.*
3. *Network graph may become too complex to visualize clearly — Focus on major routes and Top 5 airports for simplified visualization.*

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6. Sign-off

- ✓ All 5 team members reviewed and agree
- ☐ Mentor reviewed and approved
- Mentor signature / date: